Vegetable Price Prediction using Linear Regression

Abstract

In this project I tried to predict the daily price of vegetables in Indian mandis using machine learning. I used a dataset in the same format as the government's Agmarknet records (it is simulated, not real) covering five vegetables and five markets from 2022 to 2025. The main model was Linear Regression, with past prices, rolling averages, arrivals and the time of year as inputs. It predicted prices with an average error of about 25 rupees per quintal (around 1%), which is slightly better than just guessing that tomorrow's price will be the same as today's.

1. Introduction

Anyone who buys or sells vegetables knows how quickly prices can change. One week tomatoes are cheap and the next week they cost double. This happens because of things like the season, rainfall and how much produce reaches the market. Farmers and traders can lose money when they don't see this coming, so even a rough forecast can be useful.

In this project I wanted to build a simple model that predicts tomorrow's modal price of a vegetable in a particular mandi. The government publishes this kind of data on Agmarknet and data.gov.in. I chose Linear Regression as the main model because it is easy to understand and I can explain what each input is doing.

2. Dataset

I was not able to download real mandi data in bulk for this project, so I used a synthetic dataset made to look like an Agmarknet export. It has a yearly seasonal pattern, a slow upward trend, random ups and downs, and a link between arrivals and price. Since it is not real government data, the results show that my method works, not how real markets behave.

Property	Value
Rows	35,075
Period	1 Jan 2022 to 31 Dec 2025

Property	Value
Vegetables	Tomato, Onion, Potato, Brinjal, Cauliflower
Markets	Pune, Azadpur (Delhi), Bangalore, Lucknow, Kolkata
Columns	State, District, Market, Commodity, Variety, Arrival_Date, Min Price, Max Price, Modal Price, Arrivals_in_Tonnes
Missing days	About 4%, removed at random to mimic real reporting gaps

The thing I am predicting is the modal price, which is the price at which most of the trading happened that day. The same code should work on a real CSV from data.gov.in if I swap the file.

3. Methodology

I did the whole project in Python on Google Colab using pandas and scikit-learn. These were the steps:

1. **Cleaning the data.** I made the column names uniform and converted the dates to a proper date format. Prices were changed to numbers, and I removed rows with a missing date or a price of zero or less. Each commodity, market and date was kept only once, and any series with fewer than 60 days was dropped.
2. **Making features.** All the inputs come from earlier days (shifted by one day), so the model never sees the price it is trying to predict. I used the price 1, 7 and 14 days ago; the 7-day average, 7-day standard deviation and 30-day average of earlier prices; yesterday's gap between the max and min price; yesterday's arrivals and their 7-day average; the time of year (sine and cosine of the day number); and the commodity and market names as one-hot columns.
3. **Splitting the data.** I trained on the first 80% of dates and tested on the last 20%. I did not shuffle the data, because that would let the model peek at the future, which is not possible in real life.
4. **Training the models.** I compared three things: a naive baseline that just says today's price equals yesterday's price, Linear Regression, and Ridge Regression (alpha 10). I checked them using MAE, RMSE, MAPE and R^2.

4. Results

Linear Regression did best out of the three on MAE, RMSE and MAPE, but it was only slightly better than the naive baseline.

Model	MAE (Rs/quintal)	RMSE (Rs/quintal)	MAPE (%)	R^2
Naive (yesterday's price)	26.019	34.465	1.035	0.998
Linear Regression	25.172	33.429	1.001	0.998
Ridge (alpha 10)	25.634	34.144	1.019	0.998

Compared to the baseline, Linear Regression reduced MAE by 3.3% (from 26.019 to 25.172) and RMSE by 3.0% (from 34.465 to 33.429). Ridge came in between the two, so the extra regularisation did not help in this case.

All three models have an R^2 of 0.998. This is because daily prices change slowly, so even yesterday's price explains almost everything. That means R^2 alone does not tell us much here, and comparing with the baseline is the better test.

The predictions for the last day of the data (31 Dec 2025) were also close to the actual values. For example, Potato in Azadpur was 2,418 in the data and the model predicted 2,439.6, and Tomato in Kolkata was 2,472 against a prediction of 2,456.5.

5. Conclusion

In this project I built a Linear Regression model that predicts next-day vegetable prices with about 1% error on average. It beat the naive baseline by roughly 3% in MAE and RMSE, so the extra features did add a little information. But since I used synthetic data, this mainly shows that my pipeline works. I cannot claim it would forecast real mandi prices this well.

6. Limitations and future work

Limitations

- The data is simulated, so real events like floods, strikes or government policy changes are not included.
- The model only predicts one day ahead, and that is the case where the naive baseline is already very strong.

- Linear Regression cannot capture non-linear effects, for example prices shooting up when supply drops below a certain level.

Future work

- Run the same notebook on real Agmarknet or data.gov.in data.
- Add weather, fuel prices, festival dates and prices from nearby markets.
- Try Random Forest and XGBoost, and use time-series cross-validation.
- Predict a week ahead, where past prices should be more useful than they are for one day ahead.